

DEVOPS - NODEJS

PROJECT DOCUMENT

GitHub: [*https://github.com/MasoodAhmadi/DevOps/tree/project*](https://github.com/MasoodAhmadi/DevOps/tree/project)

GitLab: [*https://compse140.devops-gitlab.rd.tuni.fi/masoodahmadi/devops*](https://compse140.devops-gitlab.rd.tuni.fi/masoodahmadi/devops)

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2 Project overview

This DevOps project automates software delivery pipelines, infrastructure provisioning, and monitoring to speed up deployment cycles and improve collaboration between development services.

The goal is to automate the deployment of a Nginx-based system with port 8198 and communicate with two server such us service1 and sevice2. These services have roles of UI and API that are hosted by nginx configurations. Beside these the project has test for the API which it get “/api/system-info” system information.

2.1 System requirement - Tools

Hardware

To ensure optimal performance, the system must have the following minimum hardware specifications:

- Processor: 12-core CPU
- Memory: 16GB or higher
- Storage: at least 100GB available space for the project

Software

Operating system

- Linux (Any server such as Ubuntu)
- Mac

Development tools

- Frontend: react.js
- Backend: Node.js & express.js
- Package manager: npm
- Test framework: jest.io

Additional tool

- Git: version control system
- Docker: .yml file
- LTE versions handy

3 Implementation

In this project, I implemented a web-based user interface (UI) for managing Docker containers and services using Node.js and Docker Compose. The UI allows users to interact with services running in Docker containers, such as requesting data from an API and stopping specific services. Using the docker-compose.yml configuration file, the application can manage multiple container configurations, Nginx reverse proxy, and test service. The UI communicates with the backend via HTTP requests and triggers Docker commands such as docker stop to control containers. This solution provides a continuous way for users to monitor and manage containerized applications directly from a web interface.

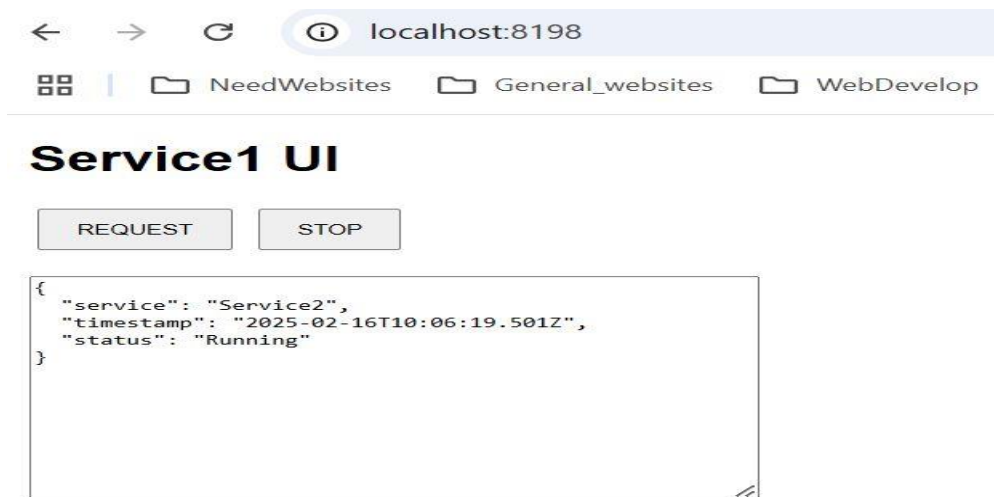
3.1 Timeframe & commits

number	Title	Description	
1	Project initialization	• Fixed exercise4 error before jumping to the project branch • Created project branch in git version control • GitHub for stashing the code • GitLab for remote repository	
2	Commits	191	

3	Pipelines	Total 386 - pass/fails	
4	Hours	77 estimations	

3.2 Pros & Cons

Services were working based on the previous exercise4 task but the docker composing file had minor mistakes and after fixing the docker-compose file it ran as you see the image below



Once the exercise4 task was completely running without error and Continuing to the implementation of project assignment there were challenges on GitLab runner, below is list of settings done after registering in GitLab-devops URL that given by teacher. <https://compse140.devops-gitlab.rd.tuni.fi/masoodahmadi/devops/-/jobs>

3.2.1 Descriptions of CI/CD GitLab

- Gitlab – settings – CI/CD
 - Runner: Assigned project runner with my environment

Runners

Runners are processes that pick up and execute CI/CD jobs for GitLab. [What is GitLab Runner?](#)

Register as many runners as you want. You can register runners as separate users, on separate servers, and on your local machine.

How do runners pick up jobs?

Runners are either:

- **active** - Available to run jobs.
- **paused** - Not available to run jobs.

Tags control which type of jobs a runner can handle. By tagging a runner, you make sure runners only handle the jobs they are equipped to run. [Learn more.](#)

Project runners

These runners are assigned to this project.

[New project runner](#)

Assigned project runners

#359 (UHYZEzN-)

[ansible](#)

[Remove runner](#)

Instance runners

These runners are available to all groups and projects.

Enable instance runners for this project



This GitLab instance does not provide any instance runners yet. Administrators can register instance runners in the admin area.

Group runners

These runners are shared across projects in this group.

Group runners can be managed with the [Runner API](#).

- Variables: store the user and password registry that is linked with the .gitlab-ci.yml file

Variables

Variables store information that you can use in job scripts. Each project can define a maximum of 8000 variables. [Learn more.](#)

Variables can be accidentally exposed in a job log, or maliciously sent to a third party server. The masked variable feature can help reduce the risk of accidentally exposing variable values, but is not a guaranteed method to prevent malicious users from accessing variables. [How can I make my variables more secure?](#)

Variables can have several attributes. [Learn more.](#)

- **Visibility:** Set the visibility level for the value. Can be visible, masked, or masked and hidden.
- **Flags**
 - **Protected:** Only exposed to protected branches or protected tags.
 - **Expanded:** Variables with **\$** will be treated as the start of a reference to another variable.

CI/CD Variables </> 2				Reveal values	Add variable
Key ↑	Value	Environments	Actions		
CL_REGISTRY_PASSWORD	*****	All (default)			
Protected Expanded					
CL_REGISTRY_USER	*****	All (default)			
Protected Expanded					

System environment:

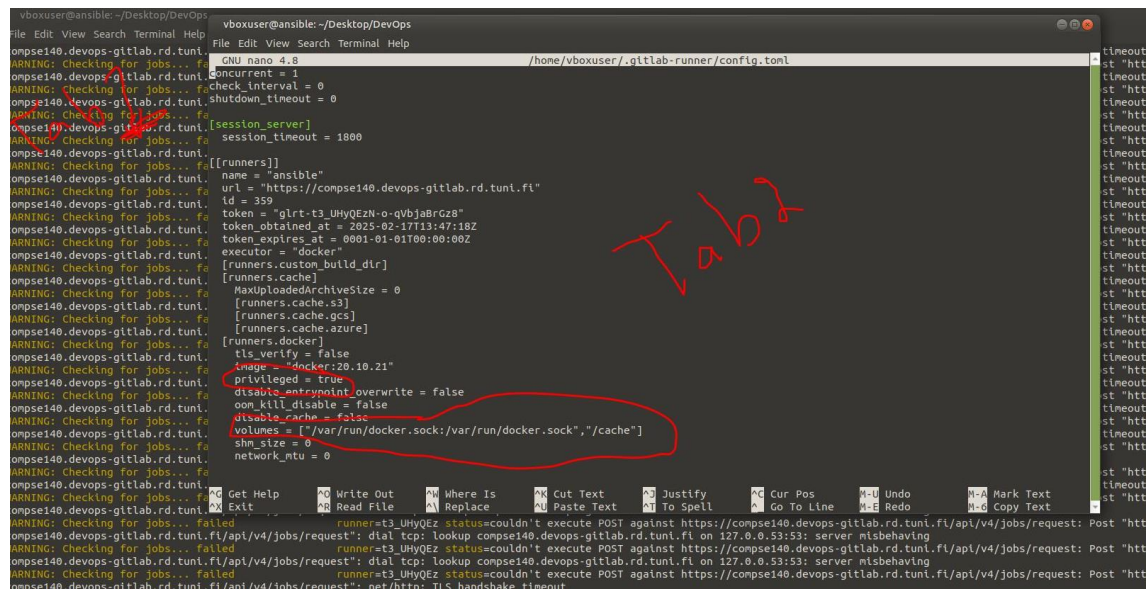
- Check in the command line that the GitLab-runner is installed, if it has not been installed then navigate to again to GitLab – CI/CD – runners and find **show runner installation and registration instructions**
- Here it should show the file that has been configured in the environment

```
vboxuser@ansible:~/Desktop/DevOps$ cat /home/vboxuser/.gitlab-runner/config.toml
concurrent = 1
check_interval = 0
shutdown_timeout = 0

[session_server]
  session_timeout = 1800

[[runners]]
  name = "ansible"
  url = "https://compse140.devops-gitlab.rd.tuni.fi"
  id = 359
  token = "glrt-t3_UHyQEzN-o-qVbjaBrGz8"
  token_obtained_at = 2025-02-17T13:47:18Z
  token_expires_at = 0001-01-01T00:00:00Z
  executor = "docker"
  [runners.custom_build_dir]
  [runners.cache]
    MaxUploadedArchiveSize = 0
  [runners.cache.s3]
  [runners.cache.gcs]
  [runners.cache.azure]
  [runners.docker]
    tls_verify = false
    image = "docker:20.10.21"
    privileged = true
    disable_entrypoint_overwrite = false
    oom_kill_disable = false
    disable_cache = false
    volumes = ["/var/run/docker.sock:/var/run/docker.sock", "/cache"]
    shm_size = 0
    network_mtu = 0
```

Continuing to this topic, there are multiple changes in this file because I faced errors in the environment setup but it might be different than python based project. The changes I have made are in the picture below:



- After the changes it should be running in a command line terminal when pushing the changes in GitLab so can communicate with each other.

```
vboxuser@ansible:~$ gitlab-runner run
Runtime platform arch=amd64 os=linux pid=5284 revision=690ce25c version=17.8.3
Starting multi-runner from /home/vboxuser/.gitlab-runner/config.toml... builds=0 max_builds=0
WARNING: Running in user-mode.
WARNING: Use sudo for system-mode:
WARNING: $ sudo gitlab-runner...

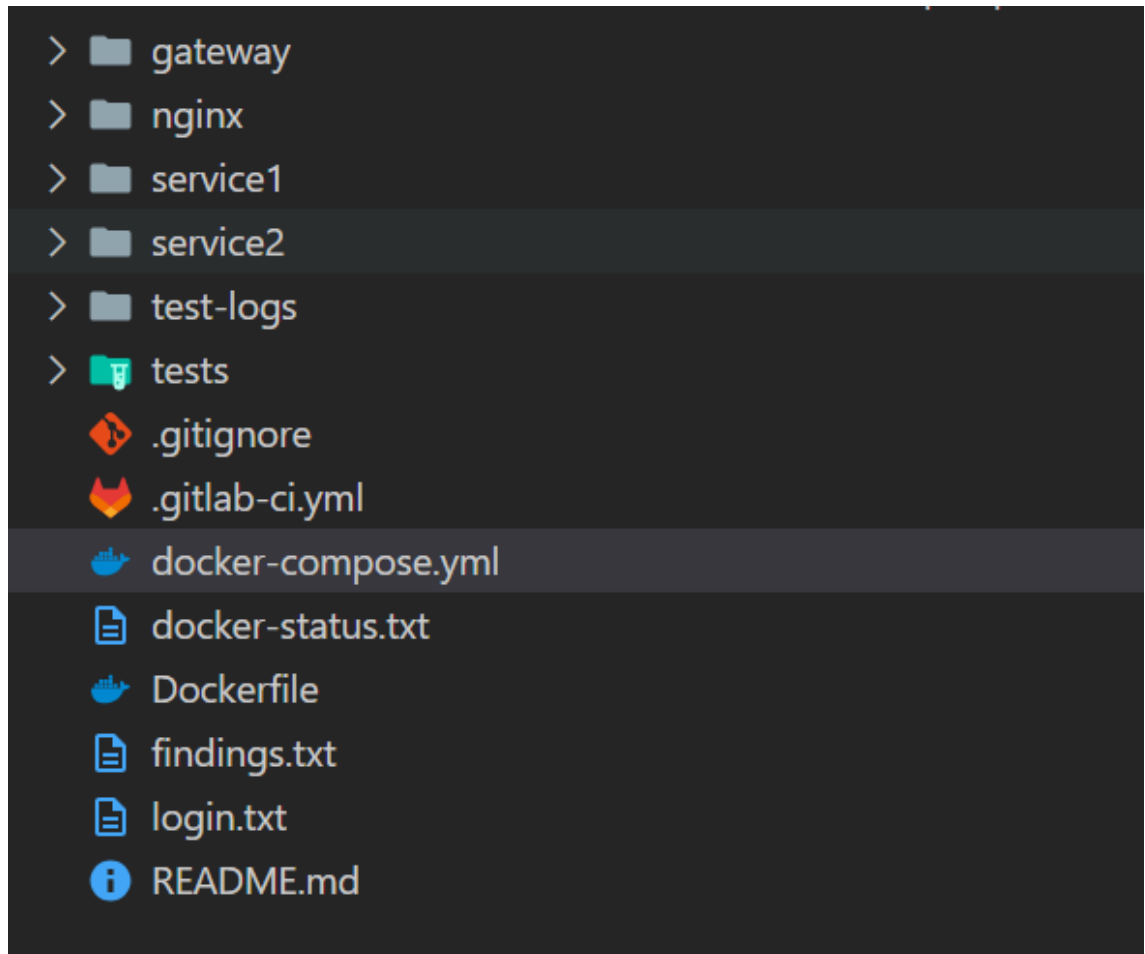
Configuration loaded
listen address not defined, metrics & debug endpoints disabled builds=0 max_builds=1
[session_server].listen_address not defined, session endpoints disabled builds=0 max_builds=1
Initializing executor providers
Checking for jobs... received
Added job to processing list
Appending trace to coordinator...ok

job=14486 repo_url=https://compse140.devops-gitlab.rd.tunt.fi/masoodahmadi/devops.git
builds=1 job=14486 max_builds=1 project=104 repo_url=https://compse140.devops-gitlab
code=202 job=14486 job-log=0-274 job-status=running runner=t3_UHyQEz sent-log=0-273
```

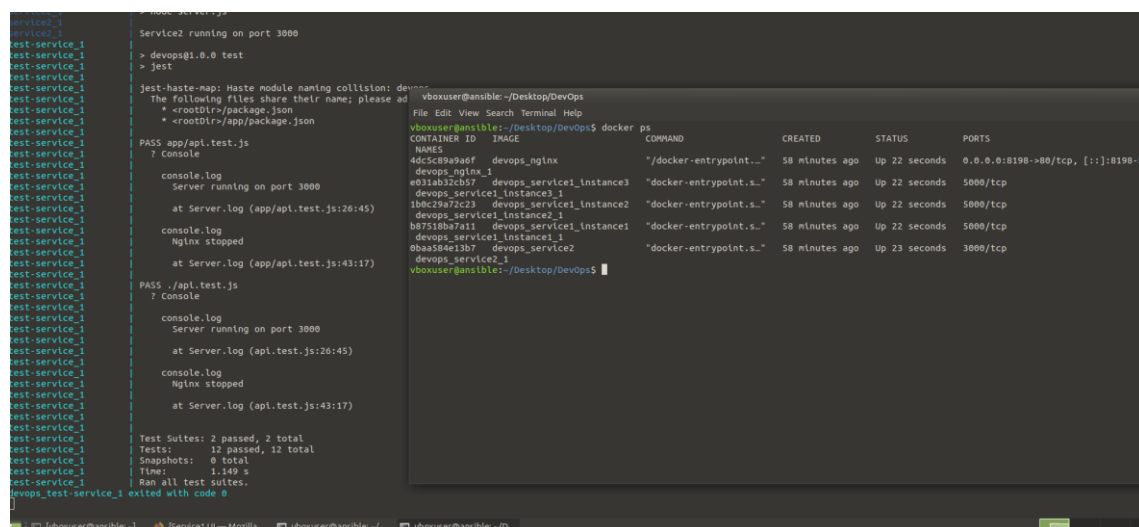
3.3 Structure

This project uses Docker Compose to manage multiple services in a containerized environment efficiently. It consists of several microservices, including service1 (with three instances), service2, nginx, and a test service, all connected to a common Docker network (devops_network).

- Service1: Three instances of service1 are deployed, all exposing port 5000, allowing for load balancing and scalability. One instance has additional Docker socket access, presumably for managing other containers.
- Service2: A separate service running on port 3000, which can act as a backup or support microservice.
- Nginx: Acts as a reverse proxy and load balancer, forwarding traffic to the appropriate service. It maps host port 8198 to container port 80, uses volumes for configuration management, and depends on all core services.
- Test-Service: Designed for automated testing, it runs after nginx finishes, and logs are continuously stored via volumes.



Docker: all the docker containers are running locally before pushing the changes into GitHub & GitLab remote project branch as shown in the picture below:



There are two git pushes that go to GitHub & GitLab

```
vboxuser@ansible:~/Desktop/DevOps$ git add .
vboxuser@ansible:~/Desktop/DevOps$ git commit -m'stop containers are working with stop button is click in the UI'
[project 8a16ca1] stop containers are working with stop button is click in the UI
7 files changed, 29 insertions(+), 21 deletions(-)
vboxuser@ansible:~/Desktop/DevOps$ git push origin project
Enumerating objects: 23, done.
Counting objects: 100% (23/23), done.
Delta compression using up to 2 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (12/12), 1.59 KiB | 813.00 KiB/s, done.
Total 12 (delta 9), reused 0 (delta 0)
remote: Resolving deltas: 100% (9/9), completed with 9 local objects.
To github.com:MasoodAhmadi/DevOps.git
6e27a32..8a16ca1 project -> project
vboxuser@ansible:~/Desktop/DevOps$ git push gitlab project
Enumerating objects: 23, done.
Counting objects: 100% (23/23), done.
Delta compression using up to 2 threads
Compressing objects: 100% (12/12), done.
Writing objects: 100% (12/12), 1.59 KiB | 1.59 MiB/s, done.
Total 12 (delta 9), reused 0 (delta 0)
remote:
remote: View merge request for project:
remote: https://compse140.devops-gitlab.rd.tuni.fi/masoodahmadi/devops/-/merge_requests/1
remote:
To compse140.devops-gitlab.rd.tuni.fi:masoodahmadi/devops.git
6e27a32..8a16ca1 project -> project
```

3.4 Testing

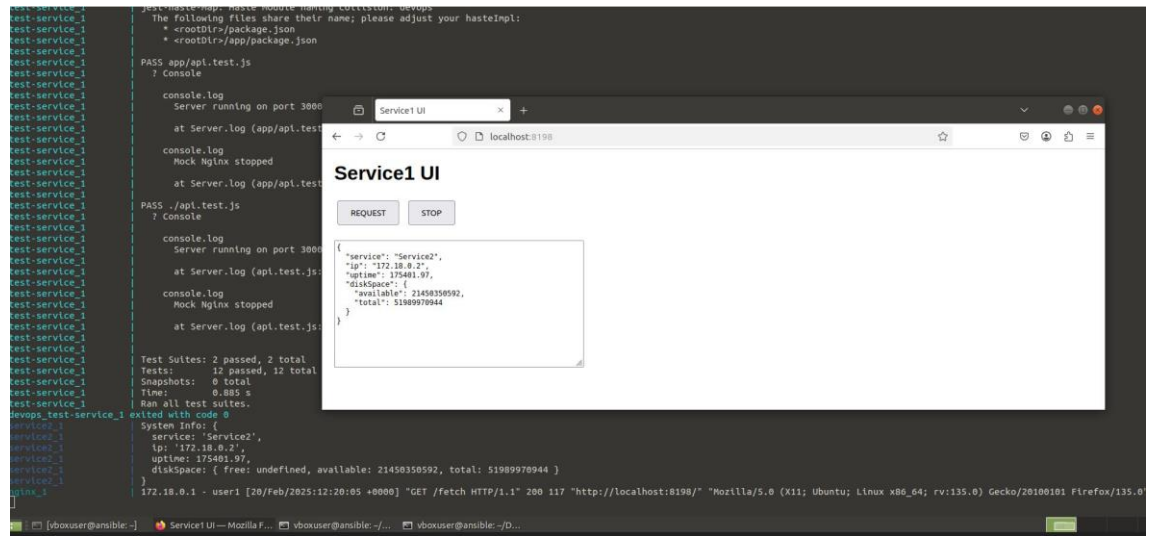
The idea of node.js testing comes from the Jest framework. It allows developers to write unit and integration tests for JavaScript applications easily. Dockerize the application with a docker-compose.yml file because it automates the testing environment and defines jest.js as a service in the container. As defined in the .gitlab-ci.yml file, the tests are automatically run every time the code is pushed to the project branch. This ensures that the code is thoroughly tested before merging, improving the quality and consistency of the code.

```
▷ Run Service
test-service:
  build:
    context: ./tests
    dockerfile: Dockerfile
  depends_on:
    - nginx
  environment:
    - API_URL=http://nginx/fetch
  volumes:
    - ./tests:/app
    - ./test-logs:/app/test-logs
  networks:
    - devops_network
```

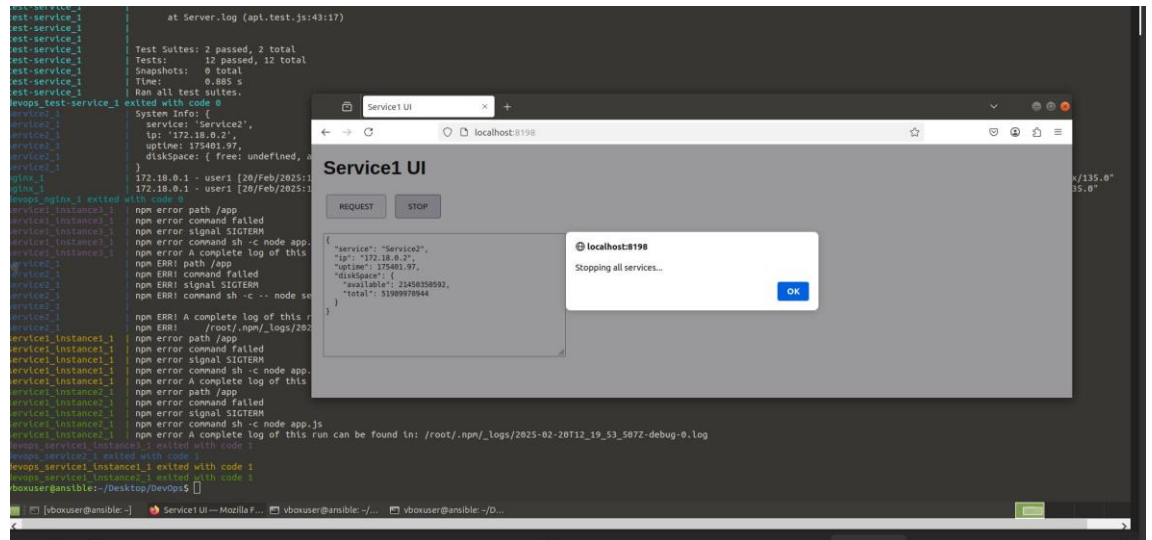
All 388 Finished			
Filter jobs			
Status	Job	Pipeline	Coverage
1 2 ✓ Passed 00:01:24 just now	#14698: test ✓ project 8a16ca1b ansible	#4329 created by Stage: test	
✓ Passed 00:03:11 1 minute ago	#14697: build ✓ project 8a16ca1b ansible	#4329 created by Stage: build	
✓ Passed 00:00:55 2 hours ago	#14696: test ✓ project 6e27a326 ansible	#4328 created by Stage: test	
✓ Passed 00:00:50 2 hours ago	#14695: build ✓ project 6e27a326 ansible	#4328 created by Stage: build	
✓ Passed 00:00:48 2 hours ago	#14694: test ✓ project 8f366af8 ansible	#4327 created by Stage: test	
✓ Passed 00:00:44 2 hours ago	#14693: build ✓ project 8f366af8 ansible	#4327 created by Stage: build	

3.5 result

- running all the containers



- stop is shutdown all containers



4 Error handling

There were many errors which couldn't build the project on the `.gitlab-ci.yml`

4.1 debugging

```
30 Pulling docker image docker:20.10.21 ...
31 Using docker image sha256:81a4721a045e389ed8754844971b3277b59757fe75c69120e6cea8dbc8de6301 for docker:20.10.21 with digest docker@sha256:812ff1098c97878ae
32 26ca8e0423e0aa157596a8b28d4bd52c120c4d087e8ae1 ...
33 Preparing environment
34 Running on runner-t3iz5qjf-project-104-concurrent-0 via ansible...
35 Getting source from Git repository
36 Fetching changes with git depth set to 20...
37 Reinitialized existing Git repository in /builds/masoodahmadi/devops/.git/
38 Checking out 3e113c1 as detached HEAD (ref is project)...
39 Skipping Git submodules setup
40 Executing "step_script" stage of the job script
41 Using docker image sha256:81a4721a045e389ed8754844971b3277b59757fe75c69120e6cea8dbc8de6301 for docker:20.10.21 with digest docker@sha256:812ff1098c97878ae
42 26ca8e0423e0aa157596a8b28d4bd52c120c4d087e8ae1 ...
43 $ docker login -u "$CI_REGISTRY_USER" -p "$CI_REGISTRY_PASSWORD" $CI_REGISTRY
44 WARNING! Using --password via the CLI is insecure. Use --password-stdin.
45 error during connect: Post "http://docker:2375/v1.24/auth": dial tcp: lookup docker on 130.230.224.10:53: no such host
46 Cleaning up project directory and file based variables
47 ERROR: Job failed: exit code 1
```

```
62 drwxrwxrwx 2 root root 4.0K Feb 17 13:20 tests
63 $ echo "Building the Docker images using docker-compose..."
64 Building the Docker images using docker-compose...
65 $ docker-compose -f docker-compose.yml build
66 no valid drivers found: error during connect: Get "http://docker:2375/v1.24/info": dial tcp: lookup docker on 130.230.224.10:53: no such host
67 Cleaning up project directory and file based variables
68 ERROR: Job failed: exit code 17
```

```
34 Getting source from Git repository
35 Fetching changes with git depth set to 20...
36 Reinitialized existing Git repository in /builds/masoodahmadi/devops/.git/
37 Checking out 3e113c1 as detached HEAD (ref is project)...
38 Skipping Git submodules setup
39 Executing "step_script" stage of the job script
40 Using docker image sha256:81a4721a045e389ed8754844971b3277b59757fe75c69120e6cea8dbc8de6301 for docker:20.10.21 with digest docker@sha256:812ff1098c97878ae
41 26ca8e0423e0aa157596a8b28d4bd52c120c4d087e8ae1 ...
42 $ docker login -u "$CI_REGISTRY_USER" -p "$CI_REGISTRY_PASSWORD" $CI_REGISTRY
43 WARNING! Using --password via the CLI is insecure. Use --password-stdin.
44 error during connect: Post "http://docker:2375/v1.24/auth": dial tcp: lookup docker on 130.230.224.10:53: no such host
45 Cleaning up project directory and file based variables
46 ERROR: Job failed: exit code 1
```

```
4928 buildx: Docker Buildx (Docker Inc., v0.9.1)
4929 compose: Docker Compose (Docker Inc., v2.14.0)
4930 Server:
4931 ERROR: error during connect: Get "http://docker:2375/v1.24/info": dial tcp: lookup docker on 130.230.224.10:53: no such host
4932 errors pretty printing info
4933 Client:
4934 Context: default
4935 Debug Mode: false
4936 Plugins:
4937 buildx: Docker Buildx (Docker Inc., v0.9.1)
4938 compose: Docker Compose (Docker Inc., v2.14.0)
4939 Server:
4940 ERROR: error during connect: Get "http://docker:2375/v1.24/info": dial tcp: lookup docker on 130.230.224.10:53: no such host
4941 errors pretty printing info
4942 Client:
4943 Context: default
4944 Debug Mode: false
4945 Plugins:
4946 buildx: Docker Buildx (Docker Inc., v0.9.1)
4947 compose: Docker Compose (Docker Inc., v2.14.0)
4948 Server:
4949 ERROR: error during connect: Get "http://docker:2375/v1.24/info": dial tcp: lookup docker on 130.230.224.10:53: no such host
4950 errors pretty printing info
4951 Terminated
4952 WARNING: after_script failed, but job will continue unaffected: context canceled
4953 ERROR: Job failed (system failure): aborted: terminated
```

```
24 Preparing environment 00:01
25 Running on runner-t3tyqjv-project-104-concurrent-0 via ansible...
26 Getting source from Git repository 00:02
27 Fetching changes with git depth set to 20...
28 Reinitialized existing Git repository in /builds/masoodahmadi/devops/.git/
29 Checking out f8bb4c06 as detached HEAD (ref is project)...
30 Skipping Git submodules setup
31 Executing "step_script" stage of the job script 00:01
32 Using docker image sha256:7015f2c475d511a251955877c2862016a4042512ba625ed905e69202f87e1a21 for docker:24.0.5 with digest docker
@sha256:3c6e4dca7a63c9a32a4e00da40461ce067f255987ccc9721cf18ffa087bcd1ef ...
33 $ docker-compose build
34 Cannot connect to the Docker daemon at tcp://127.0.0.1:2375. Is the docker daemon running?
35 Cleaning up project directory and file based variables 00:02
36 ERROR: Job failed: exit code 1
```

4.2 Resolves

To understand how to build and test successfully, please go to the GitLab setup topics and read carefully.

Masood Ahmadi / devops / Jobs

All 388 Finished			
Filter jobs			
Status	Job	Pipeline	Coverage
Passed 00:01:24 19 hours ago	#14698: test project 8a16ca1b ansible	#4329 created by Stage: test	
Passed 00:03:11 19 hours ago	#14697: build project 8a16ca1b ansible	#4329 created by Stage: build	
Passed 00:00:55 22 hours ago	#14696: test project 6e27a326 ansible	#4328 created by Stage: test	
Passed 00:00:50 22 hours ago	#14695: build project 6e27a326 ansible	#4328 created by Stage: build	
Passed 00:00:48 22 hours ago	#14694: test project 8f366af8 ansible	#4327 created by Stage: test	
Passed	#14693: build	#4327 created by	

5 Summary

The services when run with docker-compose up --build , it builds and starts all services including service1, service2, nginx, and a test service. The system fetches data from service2 via the /api/system-info endpoint, while a request button updates the API for new requests for information. Additionally, a stop button allows you to shut down all running containers. This configuration ensures seamless service communication and automated deployment.

```
vboxuser@ansible:~/Desktop/DevOps$ curl localhost:8197/run-log
2025-02-21T11:48:02.723Z: INIT -> PAUSED
```

```
vboxuser@ansible:~/Desktop/DevOps$ curl locstate -X PUT -d "INIT" -H "Content-Type: text/plain"
state changed to INITvboxuser@ansible:~/Desktop/DevOps$ curl localhost:8197/state -X PUT -d "SHUTDOWN" -H "Content-Type: text/plain"
shutting down all services...vboxuser@ansible:~/Desktop/DevOps$
```

```
vboxuser@ansible:~/Desktop/DevOps$ curl localhost:8197/run-log
2025-02-21T11:48:02.723Z: INIT -> PAUSED
2025-02-21T11:53:26.424Z: PAUSED -> INIT
2025-02-21T11:53:37.543Z: INIT -> SHUTDOWN
vboxuser@ansible:~/Desktop/DevOps$
```

6 References

Docker, Inc. (n.d.). *Docker: Empowering App Development for Developers*. Retrieved from <https://www.docker.com>

React: A JavaScript Library for Building User Interfaces. Retrieved from <https://react.dev>

Express.js Foundation. (n.d.). *Express - Fast, Unopinionated, Minimalist Web Framework for Node.js*. Retrieved from <https://expressjs.com>

Nginx, Inc. (n.d.). *NGINX: High-Performance Load Balancer, Web Server & Reverse Proxy*. Retrieved from <https://www.nginx.com>

OpenAI. (2023). *ChatGPT: Large Language Model for Conversational AI*. Retrieved from <https://openai.com/chatgpt>